2.47 Bivariate Data: Spearman's Rank Hypothesis Testing

Worked Solutions

1 a)

x	88	85	90	78	92	87	84	80	93	76	89
y	97	89	95	84	91	88	82	79	96	81	90
Rank x	7	5	9	2	10	6	4	3	11	1	8
Rank y	11	6	9	4	8	5	3	1	10	2	7
d	-4	-1	0	-2	2	1	1	2	1	-1	1
d^2	16	1	0	4	4	1	1	4	1	1	1

$$\sum d_i^2 = 16 + 1 + 0 + 4 + 4 + 1 + 1 + 4 + 1 + 1 + 1 = 34$$

$$n = 11$$

$$r_s = 1 - \frac{6 \times 34}{11(11^2 - 1)} = \frac{93}{110} = 0.845 \text{ to 3sf}$$

1 b)

Reject H_0. There is sufficient evidence to suggest that there is a positive association between Mathematics and English test scores in the population.

2)

H_0: There is no association between the length of a bird's wingspan and its tendency to migrate long distances in the population.

H_1: There is a positive association between the length of a bird's wingspan and its tendency to migrate long distances in the population.

$$r_s = 0.479$$

The critical value for $n = 12$ for a one-tailed test at a 5% significance level is 0.5035

$$0.479 < 0.5035$$

Fail to reject H_0. There is insufficient evidence to suggest that there is a positive association between the length of a bird's wingspan and its tendency to migrate long distances in the population.

3)

H_0: There is no association between the air quality index and the average life expectancy of people in different cities in the population.

H_1: There is a negative association between the air quality index and the average life expectancy of people in different cities in the population.

$$r_s = -0.654$$

The critical value for $n = 15$ for a one-tailed test at a 1% significance level is 0.6036

$$-0.654 < -0.6036$$

Reject H_0. There is sufficient evidence to suggest that there is a negative association between the air quality index and the average life expectancy of people in different cities in the population.

4)

H_0: There is no association between the number of hours of sleep a person gets per night and their cognitive performance score on a standardised test in the population.

H_1: There is some association between the number of hours of sleep a person gets per night and their cognitive performance score on a standardised test in the population.

$$r_s = 0.321$$

The critical value for $n = 22$ for a two-tailed test at a 10% significance level is 0.3608

$$0.321 < 0.3608$$

Fail to reject H_0. There is insufficient evidence to suggest that there is some association between the number of hours of sleep a person gets per night and their cognitive performance score on a standardised test in the population.